



Engineering Project

B

Semester 2 - 2025

MECH4841 A

PROJECT TITLE

NU Racing Team Leader

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FINAL YEAR PROJECT

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FYP Part B Report

NU Racing Team Leader

Cooper Chomicki



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2025

Abstract

This report details the experience and responsibilities of the author within the role of NU Racing Team Leader in 2025, highlighting the complexity and intricacies of running the University of Newcastle's Formula SAE-A team. The Formula SAE-A (FSAE-A) competition challenges university students to design, build, and race open wheeled electric vehicles, with an important focus on developing engineering practices as well as project management skills. Efforts to encourage chemistry and a positive culture have historically been proven to improve team success, where a structured leadership team further continued this trend. The team leader oversees technical documentation regarding the competition, team direction, budgeting, safe operating procedures as well as risk management.

The team leader is a holistic role, where the student has maintained the important relationships between the university, competition organisers, industry partners as well as the public. Managing conflict, human relations, timelines, as well as foresight for future teams are all elements of the team leader role which the author has continued to maintain and develop. The leadership structure provides stability and reliability within the team, allowing for the implementation of new development strategies. The successful implementation of four parallel projects, as well as continual innovation of the competition platform, are a testament to the leadership strategy.

A crucial point of focus in 2025 for the team leader was the development of junior members, as well as leadership mentoring. During the project, the author has pushed to mentor new students to decrease knowledge deficits, and in turn increase knowledge transfer. A result of this strategy was selecting the 2026 team leader in August, therefore further enhancing potential as well as building solid relationships earlier than ever before. The team leader also helps to facilitate the development of high-quality engineers, which not only benefits NU Racing, but the University of Newcastle. This development is completed through extensive project reviews, appropriate resourcing as well as working closely with academics.

In 2025, NU Racing achieved its first podium at the Formula SAE-A event in its 23-year history. This reward is a testament to not only the excellent group of engineers who played a part in NU25, but the complex leadership strategy and relentless pursuit of performative excellence. NU Racing netted six trophies, including the technical inspection award, a testament to the team's preparedness and safety. The author's push for performance through a productive work environment as well as a data-driven decision-making resulted in the best finish in competition history for NU Racing.

The leadership experience highlights the complexity and reward of managing a student-run motorsport team, emphasizing the importance of organisation, morale, and continuous improvement in pursuit of competitive success.

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1. Introduction

Formula SAE A (FSAE-A) is a competition where “university students design, build, and compete in their very own Formula-style open wheeled racing car.” [1]. This competition gives a platform to a huge array of students, to showcase their engineering ability, as well as project management skills. Each university competing within FSAE-A has a ‘Team Leader’, a person who is tasked with managing their respective teams entirely. This role is a large responsibility, and its difficulty is often underestimated.

The author of this report accepted the role of NU Racing Team Leader in December of 2024. This role is one of crucial importance, whereby the author accepted responsibility to not only lead the team, but to push for performance and reliability. The role of Team Leader entails a variety of different tasks, largely being a project management-based role. This means the Team Leader is tasked with guiding a Formula Student Racing team of 50+ people throughout the course of the year, navigating problems, setbacks, and more importantly, celebrating wins. The goal of NU Racing Team Leader is to handle all administrative duties, meetings, student conflict, interacting with Industry Partners as well as push their team forwards, striving for excellence, reliability and ultimately a podium at the FSAE-A competition.

Leadership within NU Racing has been much the same for the previous years, building on a structured system allowing for transparent and effective leadership. Below is an organisational chart outlining the leadership system within NU Racing, as well as the students within those roles. Importantly, the leadership team is comprised of Team Leader, Chief Engineer, and Department Leads for both Mechanical, and Mechatronics Systems. Within this team, an academic supervisor is also required, allowing for more depth and insight when managing a team. Great leaders know that they can’t do everything on their own and they’re good at finding the right people to help them lead.

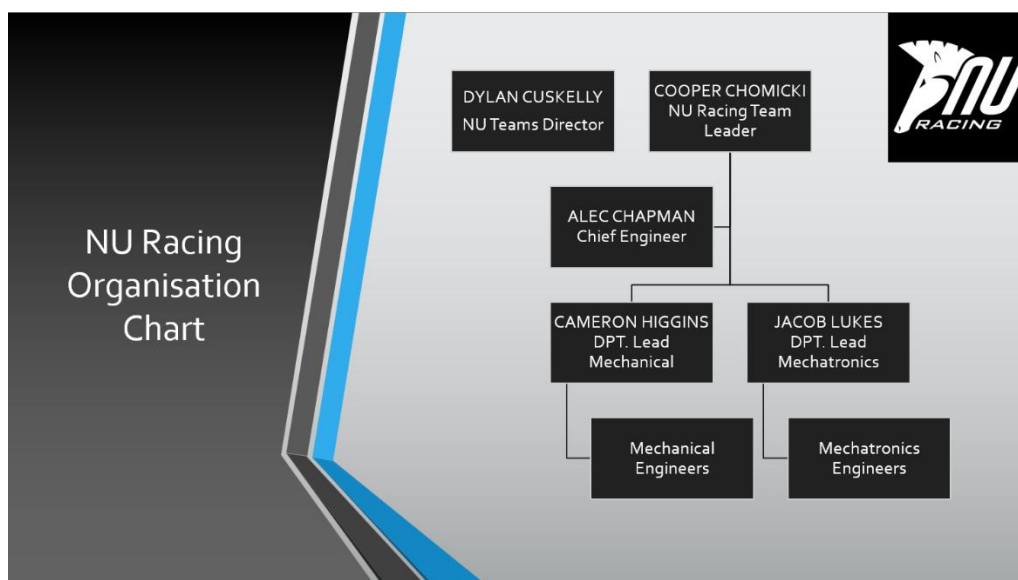


Figure 1 - NU Racing Organisational Chart

Over the course of several months, the author has settled into their role, implementing new and exciting ideas, as well as continuing previous tradition within the role. Since the commencement of the position, the author has implemented several new documents, such as the 2025 NU Racing budget, project plan (Gantt chart), attendance forms, weekly progress log, as well as a Guide to Using University Vehicles. These documents allow for the smooth operation of a university student run racing team, highlighting the importance of organisation and preparation. Excerpts of these documents can be found in the Appendix as well as throughout this report.

This document will elaborate on the progress of the author within their role, detailing the finer aspects of NU Racing Team Leader.

2. NU Racing Team Leader Scope & Roles

The Team Leader role within NU Racing is important and entails many different responsibilities. These were outlined at the beginning of the year when the leadership team established the 2025 NU Racing FYP & Directed Roles [2] document, which contains all scopes and roles of each member within the team.

The problem statement of the NU Racing Team Leader is defined as the following:

“To lead & manage NU Racing in its entirety. This involves managing 18 Final Year Project students, 14 Directed Reading students, Extra Curricular students, conducting leadership tasks, task delegation, as well as managing upcoming events. This also involves managing the risk of team activities, ensuring the safety of all participants at all times.

The role requires working closely with NU Teams Academic Supervisor, Dr Dylan Cuskelly, to convey the overarching goals of NU Teams within NU Racing, as well as continue fostering a productive and performance-based environment within the team. The team leader is tasked with managing budget, attendance of team members, establishing scopes and roles, as well as providing students the opportunity to succeed given their projects by providing available and adequate resources.”

3. Managing a Formula Student Racing Team

Most notably, the author is continually managing the entirety of the NU Racing team. This task is daunting and constantly evolving, meaning the organisational skills of the author are always tested. NU Racing is comprised of 50+ students, where the leadership team is tasked with managing all students within their respective roles.

Task delegation is a large part of the Team Leader role, as the sheer size and complexity of the team means without an effective project management plan, the programme will fall behind. Delegating tasks to the Chief Engineer, as well as MECH/MCHA Department Leads was a necessary step in the smooth operation of the team this semester.

3.1 Leadership Selection

A notable decision in 2025 was the confirmation of the 2026 leadership team in October, the earliest in NU Racing history. This organisational push has made a huge impact on the team, allowing new leadership members to become familiar with processes as well as gain the necessary knowledge transfer from their predecessors. This knowledge transfer has historically been a large problem within NU Racing, where outgoing students take a lot of valuable knowledge and resources with them. One student who was a part of this process stated “being selected early inspired confidence in me to be able to gain knowledge of the car & team as early as I could. It has allowed me to grow and adapt my understanding before undertaking the role.” The decision to elect early and mentor new leadership students coming through the programme is one in which NU Racing will greatly benefit from in 2026.

3.2 Communicating with a team

The author continued the use of the NU Teams Discord in 25', centralising all official communication with the team. Split channels allow for directed communication, as well as streamlining important information on mainstream channels such as 'Racing Announcements'. The author published weekly announcements, summarising important information from the week in an easy-to-read post, benefitting students who may not have been in attendance for the team meeting.

As well as weekly announcements, the author used voice recordings to capture team meetings regularly, which gave students the opportunity to listen at their own pace. This was received overwhelmingly well by the team, in which the author actioned and continued to film each meeting. These meetings were published in the 'Racing Announcements' channel in Discord.

3.3 Conflict Management

A part of the team leader role is to manage all conflict within the team. As a team of young, hard-working people, conflict is bound to occur naturally in high-stress environments. The author was tasked several times with managing conflict throughout the year, most notably a major disconnect between both departments & higher end management. This was due to a variety of reasons, in which were discussed in a formal meeting between the leadership team. This meeting resulted in a whiteboard full of feedback, in which the author delivered in front of the entire team.

Leadership accountability is a critical part of being a good leader, something the author focused on heavily throughout the year. This meeting resulted in a variety of actionable changes, most affecting the way the author lead the team and the level of commitment demonstrated. These changes were swiftly implemented and were received well by all, highlighting the accountability and desire to be a better leader.

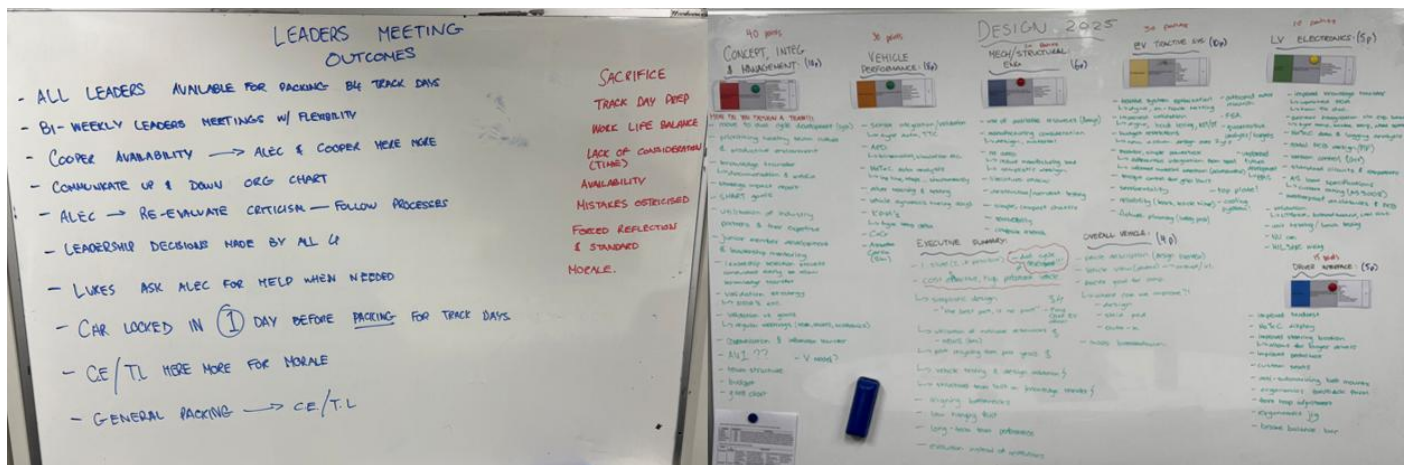


Figure 2 - Leadership Whiteboards 2025

The right whiteboard in Figure 2 - Leadership Whiteboards 2025 demonstrates another aspect of leadership, highlighting the preparation carried out by the author in collaboration with the Chief Engineer, Alec Chapman, for the Design event at competition. This model of brainstorming was a regular occurrence throughout 2025, allowing ideas to be visualised and cross examined, creating robust & meaningful decisions.

4. Sustainable Growth of NU Racing

A key element to the authors role is to ensure the positive growth of the programme throughout the year, where success in the future is created in the present. Onboarding new students was crucial for 2025, where the leadership team successfully gained 18 new Final Year Project (FYP) students. This group was the foundation for the year, where all initial projects were allocated and work began on the competition platform, as well as several parallel projects.

4.1 Junior Onboarding

All students that join NU Racing are onboarded through a series of small, task-based learning modules called 'Zero-to-Heros'. This onboarding process allows all students to have a common baseline, which is the foundation for all things done within the team. In 2025, the author also took initiative to change the way junior students are welcomed into the team. Typically, it is the team leader's responsibility to manage all new junior students, although due to the large workload of the competition platform, some students are left disadvantaged or without the support they need. The author identified this issue and implemented a revolutionary project to ensure the smooth running of 'ENGG2200', the 'Intro to NU Teams' course. NU Racing in

2025 resourced an 'ENGG2200 Technical Advisor', in which a senior student, Sam Smythe, was able to manage a large portion of responsibility within the junior onboarding process.

This project was a huge success, reducing the stress and workload of the team leader, allowing the author to focus on the competition platform and ensure its success at competition. The author also valued the importance of staying connected with the new students however, ensuring to regularly help where needed as well as being involved in events such as demonstration day. 23 Students successfully completed ENGG2200 in 2025, in which 6 of attended the annual competition well after course completion. These 6 students played crucial roles for the team at competition, highlighting the importance of junior onboarding and the value of adequate resourcing to new students.



Figure 3 - Junior Demonstration Day 2025

4.2 Sponsorship & Industry Partnership

The author pushed hard to grow exposure for NU Racing in 2025, focusing on reaching out to potential partners, as well as boosting social media usage to promote engagement. During Semester 1, NU Racing posted 4.8 times more compared to that of 2024, keeping our community engaged as well as showing potential partners what NU Racing has to offer. As of November 2025, the last 5 posts on the NU Racing Instagram have averaged 19.5k views, pushing our social media to the next level.

This positive push with the teams marketing has led to an increase in support, in which five new sponsors have partnered with NU Racing in 2025. As well as an overall increase in sponsorship, the author in collaboration with the 2026 Team Leader, Holly Nicholas, has begun to establish a development pathway with Supercars, the governing body of the highest level of motorsport in Australia. This partnership has potential to take NU Racing to the next level, as this is something that no other Formula SAE programme can offer. This pathway would involve undergraduate students within not only NU Racing, but NU Teams, and would grant opportunities in the motorsport industry by working on real world projects, through Supercars and their partners. These projects give undergraduates motorsport related industry experience, exposing them to high-pressure environments as well as a taste of professional motorsport.

In 2025, with the help of alumni student Joshua Hayward, NU Racing also shot an official reveal video for the competition platform, NU25. This video was published on all platforms, and was received extremely well by industry, sparking a new interest in NU Racing. This video showcased not only the exceptional engineering involved in the construction of NU25, but also the partners and sponsors that facilitated another year of pushing the boundary of what NU Racing can do.

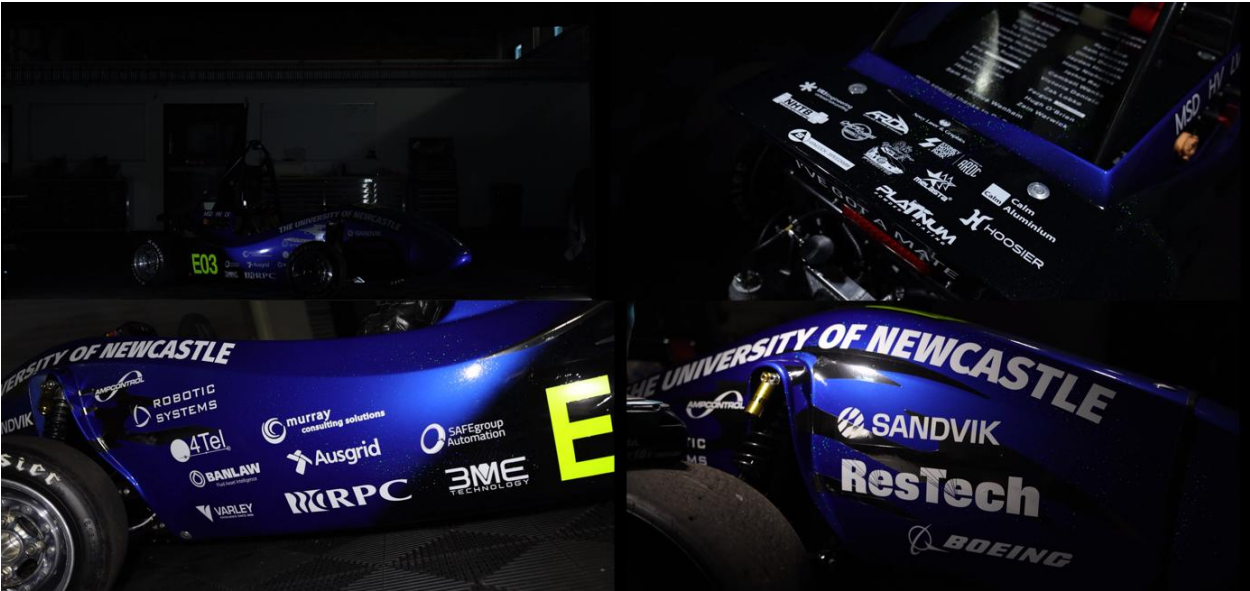


Figure 4 - NU25 Reveal Video

5. Leadership Structure

NU Racing is a team which historically prides itself on preparedness, commitment & a willingness to learn. These positive aspects of culture are all further cultivated through leadership structure, where the author implemented a variety of exciting ideologies to continue boosting this culture.

5.1 Meetings & Scheduling

An important document that was introduced by the author was the ‘Weekly Progress Log’, in which students completed each week prior to the team meeting.

NU Racing Weekly Progress Log		Meeting Date	What I did this week	What will I do next week	What challenges/risks did I face this week	Did I ask for help if I needed it
Fill in each week before the Wednesday 2-3pm team meeting. Do not alter past logs if goals/deadlines are not met.		26/03/2025	Facilitated SMSP visit for 27/3, 17/4	Safety documentation for further events	Working around timelines for different reaso	Yes!!
			Organised Merchandise order	Industry Partner visit organisation	Managing personal life	
			Populated attendance form for both track days	Meet with Bill and update him on team progress		
			Updated SharePoint with new documents	Finish master transport doc		
			Organised transport to and from SMSP			
			Started master doc for car transport			
		2/04/2025	Comp Accom pencilled in	Varley Site visit	Organising transport	Yes!!
			Booked more SMSP	On Campus track day		
			Booked On Campus			
			Managed first SMSP day			
			Met with Bill			
		16/04/2025	Merch invoices distributed	Take some time off woowoo	Sticking to my schedule	Yes!!
			Cars organised			
			SMSP organised			
			Course work			
		30/04/2025	Enjoyed some time off	Attend Hunter Innovation Festival meeting	Catching up on course work	Yes!!
			Got back up to speed	SMSP track day		
			Reviewed budget			
			Organised SMSP travel for 8/5			
		28/05/2025	Directed Reading project development	Attend ENG1500/MECH2120 demos for recruitment	Report writing	Yep

Figure 5 - Weekly Progress Log

This progress log was critical for the team, keeping all students educated and up to date surrounding the current state of the programme. Student feedback was also collected by the author, validating this document and deeming it important moving forwards, where many students felt it was not only a nice way to stay connected to the team, however, to also speak to their peers about what they are currently working on. As a leader, it also holds the team accountable to continue pushing and working hard, using their team members as motivation.

As the team leader, it is critical to maintain schedule. The driving force behind preparation and organisation is accurate scheduling. During 2025, the author pushed the regular use of Outlook Calendar for all booking and scheduling, allowing academics, leaders, and students to understand commitments and plan around them. Weekly meetings between Racing leaders, team leaders, and academics allowed for scheduled checkpoints,

enabling information to be shared between groups and actioned appropriately. These meetings were the foundation for all decisions made throughout the year, as a rational thought process with a multi-pronged approach always resulted in the most appropriate outcome.

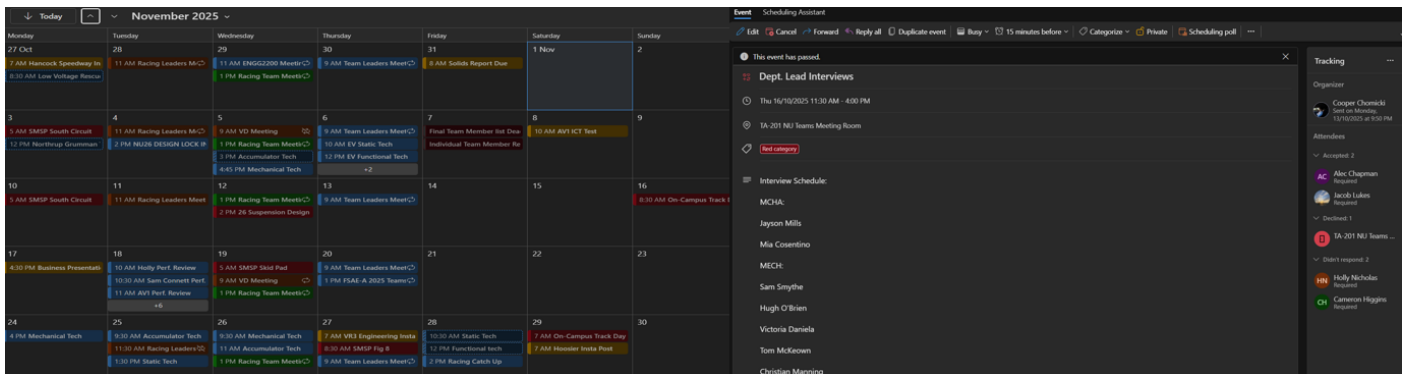


Figure 6 - Outlook Calendar & Event Scheduling

5.2 Team Chemistry

As a relatively small team in comparison throughout FSAE-A, one of NU Racing's greatest attributes is team chemistry. A huge focus for the Team Leader is to ensure the comfortability and wellbeing of all members, as this not only increases productivity, but the drive to be better. The author has focused on this aspect by organising team bonding exercises such as Go-Karts, as well as providing the appropriate setting to be positive and productive.

Another small but meaningful tradition within NU Racing is the Engineer of the Month award, where a student within the team is recognised each month for their incredible work. This year, the author has continued this trend as it is a simple yet effective way to reward students, where they receive a custom mug as a prize. This is another example of boosting team chemistry, arguably the most important aspect of NU Racing.

Due to the authors previous experience working in teams, it is evident that team chemistry is a critical element to success. It is also something that cannot be forced, therefore often making it difficult to achieve. The leadership structure within NU Racing as well as the personability of its members inherently boosts chemistry, ensuring that all team members feel comfortable and welcome. This positive team chemistry has created an excellent working environment, where students feel pushed to do better and be more productive, purely due to feeling comfortable in the NU Racing environment.

5.3 Track Days

NU Racing regularly holds track days to test different aspects of the competition platform. These track days are often held at Sydney Motorsport Park, requiring a large amount of organisation before attending.

For a NU Racing track day, students are required to arrive at university at 5 AM, to then migrate into university vehicles. These vehicles are organised by the Team Leader, where attendance needs to be calculated to allow for the right number of cars. The author regularly issues attendance forms to the team well in advance, which data is extracted from and placed into an attendance spreadsheet. This spreadsheet can be found in Appendix 8.1.

The author also must communicate with coordinating staff from the Australian Racing Drivers Club (ARDC) to book the track for testing. This process is tedious, as aligning both the SMSP track schedule as well as the NU Racing schedule becomes difficult. Currently, at the time of the curation of this document, the author has successfully held 16 Sydney Motorsport Track Days, as well as 7 On-Campus university track days. Notably, the first SMSP track day of 2025 was held with 45 students attending, all of which were transported to the track via university vehicles.

As a continuation of the sustainable growth of NU Racing, the author concentrated on getting the team on new track surfaces in 2025, as the facilities the team regularly use at SMSP can limit the data collection the team is able to complete. Throughout the year, the author successfully secured testing time on the SMSP South Circuit for the first time in NU Racing history, as well as time at the Sydney Dragway Carpark. The team completed three days on the South Circuit, and two days on the Dragway Carpark, in which the team performed flawlessly.



Figure 7 - Author driving on SMSP South Circuit

Not only was the team able to collect new and valuable data from these testing days, this opened the opportunity for a new level of driver training. This was resourced in Semester 2 to harness its potential, where a 4000-level engineering project was dedicated to driver training. The total testing time and kilometres are summarised in Table 1 - 2025 Testing Time.

Table 1 - 2025 Testing Time

<i>2025 Testing</i>	<i>NU24</i>	<i>NU25</i>	<i>Total</i>
<i>Kilometres</i>	319.1	523.3	842.4
<i>Time</i>	9hr 44min	16hr 07min	25hr 51min

This continual push for growth is an aspect the author relentlessly worked towards in 2025, which greatly benefitted the programme at competition in Melbourne.

5.4 Critical Decision Making

The author was responsible for many critical decisions made throughout the year, as this is an important aspect of the team leader role. This critical decision-making process is one that was refined throughout the year, after the undertaking of several literature reviews as well as learning from previous experiences. The author began exploring the use of Failure Mode Effect Analysis templates, as highlighted in Figure 8 - Failure Mode Effect Analysis (FMEA).

Failure Mode and Effects Analysis (FMEA)

RPN Columns contain a formula to auto-calculate; do not alter or delete.

ID	System / Item / Process Step	Function	Potential Failure Mode	Effects of Failure	Severity	Causes	Occurrence	Current Controls		Detection	RPN	Recommended Actions	Owner	Date Due
#	Name, ID number, etc.	Primary function	How could the system / item / process potentially fail?	Consequential impact on other systems, departments, etc.		All contributing factors		Prevention	Detection		Risk Priority Number	Steps required to reduce severity, occurrence, and detection	Organization, team, or individual responsible	Target date of completion
1	Front/Rear Control Arms	Defines camber, toe, anti-dive, anti-squat	Fatigue cracking/fracture	Sudden geometry change resulting in crash/catastrophic failure	10	Poor design (stress concentration), manufacturing defects, abnormal wear	4	Accurate FEA, material knowledge, qualified fabrication	Scheduled NDT after impacts, regular bolt checks	5	200	Lessons learnt doc, validated theoretical model with real world testing	MECH Dept.	03/31/26
2	Front/Rear Uprights	Transfer wheel loads to suspension arms, provide mounts for wheel bearings & susp. Arms	Upright fracture, bolt hole elongation	Collapsed suspension, loss of steering, results in crash/catastrophic failure	10	Impact, poor design, manufacturing issue	3	Accurate FEA, material knowledge, qualified fabrication	Inspection after impact, visual inspection, NDT	4	120	Lessons learnt doc, validated theoretical model with real world testing	MECH Dept.	04/01/26

Figure 8 - Failure Mode Effect Analysis (FMEA)

This template allowed the author, in collaboration with the leadership team, to evaluate risk on any given subsystem & determine the best way to mitigate said risk. The FMEA combines several factors, including severity, occurrence, as well as detection, to generate a risk priority number. These numbers can then be arranged to highlight a certain subsystem which is deemed 'high-risk', which can then be focused on to ensure the safety of all participants, as well as maximum performance outcome for the team without failure.

As well as the FMEA template, the author in collaboration with the Chief Engineer, Alec Chapman, generated a driver selection matrix to assist in competition driver selection. The generation of this matrix was caused due to feedback from several team members, highlighting confusion in the selection process as well as a lack of clarity surrounding 'why'.

Endurance/Efficiency										
Rate everything out of 10		Alec Chapman		Cooper Chomicki		Cameron Higgins		Driver 4		Criteria Meaning
		ENTER SOLUTION		ENTER SOLUTION		ENTER SOLUTION		ENTER SOLUTION		
Criteria	Weight	Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score	
Mass	5	10	50	6	30	8	40		0	How much does the Driver weigh compared to the others?
Consistency	10	10	100	10	100	9	90		0	From their practise, was the driver consistent? Do they miss gates and hit cones?
Experience	9	9	81	8	72	8	72		0	Has the driver driven at competition before and how much have they driven during testing?
Lap time	7	9	63	8	56	9	63		0	From testing, what drivers set the best times?
Pressure	9	9	81	9	81	8	72		0	Does the driver handle pressure well?
Adaptibility	9	10	90	10	90	10	90		0	Can the driver adapt well to different conditions?
Commitment	10	10	100	10	100	10	100		0	Does the driver attend track days and practises on the SIM
Receptive	10	10	100	10	100	10	100		0	Does the driver listen to instruction and feedback
Total Score out of:	690		665		629		627		0	

Figure 9 - Competition Driver Selection Matrix

This matrix was generated for each competition event, highlighting important criteria to the leadership team. This was used as a tool to assist in driver selection, however, was not the sole reasoning behind the decision, as it is a complex & intricate task, using a matrix alone would have been inappropriate. This template, as well as the criteria meaning, was shared with the team via SharePoint, giving insight into the selection process as well as what criteria the decision was considering.

This not only allowed for more clarity within the wider team, but also within the leadership team responsible for the decision, as it is easy to cloud judgement over an important decision.

In future, this matrix should be used at the beginning of the year to assess possible drivers earlier, rather than using 'feel' to begin with. This matrix is designed to evaluate a driver at any point, with little to no experience, hence would be beneficial to the team in 2026 to use this earlier.

5.5 SharePoint

Another addition to the NU Racing team in 2025 is the use of SharePoint, an online database for all documentation used within the team. This was originally setup by the Chief Engineer, Alec Chapman, who imported all old data onto the new software, in which his work reduced once this was complete. The author is

responsible for constantly updating the database, as well as visually editing the page to make it more appealing for 100+ university students.

The SharePoint allows for a central home for all important and necessary data, as well as a page to keep the team updated. Several features have been explored within the software, one of which is the 'Events' section. This allowed the author to enter all upcoming track days and important events, allowing the team to stay organised and increase attendance.

Having a centralised database allows for consistent direction for the team, where all documents required can be found within the SharePoint. This was a huge improvement for team communication in 2025 and should be continued into 26'.

6. Competition

NU Racing successfully attended the Formula SAE-A competition in 2025, marking a huge success for the author within the role of team leader. Competition is renowned for taking large tolls on teams, often experiencing catastrophic failure within the event. In 2025, the author took to the challenge of marking history within NU Racing, hunting for success through validation, persistence and preparation.

The team's level of preparedness, as well as extreme resilience resulted in an excellent run towards the competition, completing six track days and nine mock technical inspections in the space of twelve days. The author was tasked with not only holding a team of exhausted people together, but also transporting, feeding & housing 44 people in Melbourne, twelve hours from home. The logistical difficulty of the competition was handled with determination and confidence, where all students travelled to and from successfully.

FSAE-A 25' not only marked a safe trip, but a successful one on & off track. The team achieved its first podium in NU Racing's 23-year history and secured the largest trophy haul (six) of any team beforehand. One of many awards was the Technical Inspection award, highlighting the team's preparation, safety, & dedication to a compliant vehicle. The 25' competition is not only the most successful in history, but the ultimate reward for the years' worth of sacrifice, dedication, hard work and sheer determination. The determination to be the best team possible. Team success relies on resilience, captured perfectly in Figure 10 - NU Racing 25' FSAE-A Competition

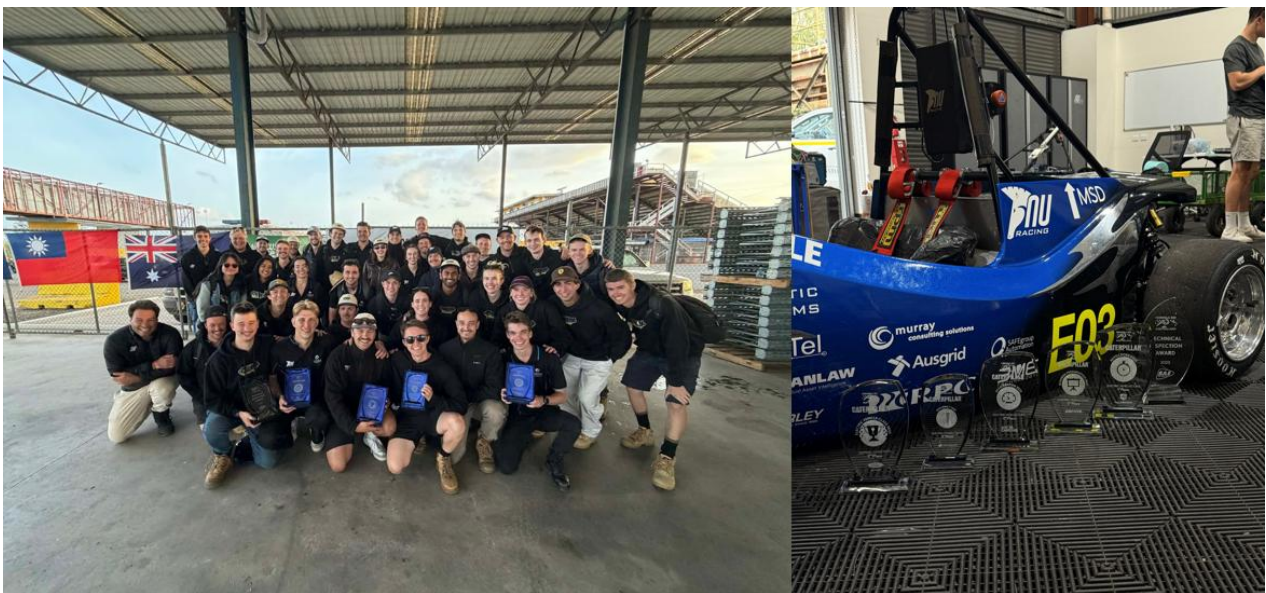


Figure 10 - NU Racing 25' FSAE-A Competition

7. Conclusion

In summary, the role of team leader within NU Racing is a holistic one, capturing a wide variety of engineering complexity, project management skills & human relations. The author has showcased a broad development of skills throughout the course of 2025, learning what it means to be a leader, as well as push for performance through validation, dedication & preparation. The implementation of documents such as Failure Mode Effect Analysis, decision matrix', & budgeting all demonstrate methodical ways of solving an engineering problem, whilst also enhancing NU Racing through early election of future leadership & junior onboarding enhancement. The author is also responsible for the most successful competition in the team's 23 year history, a testament to the leadership strategy, culture & push for performance. Overall, the safety, performance & management of NU Racing in 2025 have benefitted greatly from the authors contributions, in which the personal development & learning undertaken during the year is simply invaluable.

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9. Appendix

9.1 Attendance Spreadsheet

Track Day Attendance Planner 2025																	
Name	22-Mar	27-Mar	12-Apr	17-Apr	Varley	8-May	25-May	29-May	12-Jun	26-Jun	7-Aug	21-Aug	1-Oct	28-Oct	18-Nov	25-Nov	27-Nov
Cooper Chomici	N	Y	Y	Y	Y	Y	Y	Y	Y								
Alec Chapman	Y	Y	Y	Y	Y	Y	Y	Y	Y								
Jacob Lukes	Y	Y	Y	Y	Y	Y	Y	Y	Y								
Cameron Higgins	Y	Y	Y	Y	Y	Y	Y	Y	Y								
Brady McNeil	Y	Y	Y	Y	Y	Y	Y	Y	Y								
Cambell Wanke	N	Y	N	Y	Y	Y	Y	Y	Y								
Ethan Bailey	N	Y	Y	Y	Y	Y	Y	N	Y								
Grace Nimmo	N	Y	Y	Y	Y	Y	Y	Y	Y								
Holly Nicholas	N	Y	Y	Y	N	N											
Jayden Horvath	N	Y	Y	Y	Y	Y	N	N	Y								
Jethro Raj	N	Y	N	Y	Y	Y	N	Y	Y								
Joshua Garbutt	N	Y	Y	Y	Y	Y	N	Y	Y								
Oliver Bunney	Y	Y	Y	Y	N	Y											
Oscar Rudman	Y	Y	N/A	Y	N/A	Y	Y	Y	Y								
Ryan Bird	Y	Y	Y	Y	N	Y	Y	N	Y								
Sam Connett	Y	Y	Y	N	Y	N	Y	Y	N								
Ty Doubiklin	N	Y	N	Y	Y	Y	Y	Y	Y								
Jeffrey Keena	N/A	N/A	N/A	N/A	N/A	Y											
Jayden Hardinge	N	Y	N/A	N/A	N/A	Y	N	N	Y								
Josh Cooley	N/A	N/A	N/A	N/A	N/A	N/A											
Zain Warwick	N/A	N/A	N/A	N/A	N/A	N/A											
Oliver Dowzer	N/A	Y	N	Y	Y	N/A	Y	Y	N								
Sam Smythe	Y	N/A	N	Y	N/A	N/A	Y	N	N								
Cooper Harrod	N/A	N/A	N/A	N/A	N/A	N/A											
Mitch Jeffery	Y	Y	N/A	Y	N/A	N/A											
Victoria Daniela	N	N	N	Y	N/A	N/A	N										
Joel Evans	N/A	N/A	N/A	N/A	N/A	N/A											
Mia Cosentino	N/A	N/A	N	Y	N/A	N/A											

27/3 Track day SMSP													
Attendee	Uni Driver	Comp Driver	Experience	Column	Senior Members:	Junior Members:							
Cooper Chomici	Uni Driver		FYP										
Alec Chapman	Uni Driver		FYP		With Cars at full capacity incl.	Required 5 Seater Cars:							
Cameron Higgins	Uni Driver		FYP										
Jacob Lukes	Uni Driver		Senior										
Brady McNeil			FYP										
Cambell Wanke			FYP										
Ethan Bailey	Morisset		FYP		Truck:	Cooper Chom							
Grace Nimmo			FYP										
Holly Nicholas			FYP										
Jayden Horvath			FYP										
Jethro Raj			FYP		CHEM Hiluc:								
Joshua Garbutt	Catherine Hill Bay		FYP										
Oliver Bunney			FYP										
Oscar Rudman			FYP										
Ryan Bird			FYP										
Ty Doubiklin	Morisset		FYP										
Jayden Hardinge	Kariong		Directed		CIVIL Hiluc:								
Hugh O'Brien			EC										
Jayson Mills			Directed										
Fletcher Cogoi	Kariong		EC										
Fergus Manzie			EC		Hyundai Kona								
Zak Lobko			Directed										
Jackson Boustany			Senior										
Marisa McLean			Senior										
Tim Kerr			Retired										
Kris Kerr			Senior										
Kieran Burgess			Senior										
Josh Hayward													
Luca A Santorelli													
Christian Manning													
Jye Haller													
Mitch Jeffrey													
Johnny Jones													
Oliver Dowzer													
Harro S													

Time	Activity	Head Marshal	Marshal 2	Marshal 3	Marshal 4	In Charge
8:30	Set Up					Cooper Chomici
9:00						Cooper Chomici
9:30						Cooper Chomici
10:00						Cooper Chomici
10:30						Cooper Chomici
11:00						Cooper Chomici
11:30						Cooper Chomici
12:00						Cooper Chomici
12:30						Cooper Chomici
13:00						Cooper Chomici
13:30						Cooper Chomici
14:00						Cooper Chomici
14:30						Cooper Chomici
15:00						Cooper Chomici
15:30						Cooper Chomici
16:00						Cooper Chomici
16:30						Cooper Chomici

Driver List:	Pedal Position
Hayward	4
Kieran	4
Jayden	4
Kris	5
Jackson	5
Johnny	5 Charge!
Zak	5
Jye	5
Tim	5
Marisa	2
Holly	2 Charge!
Lukes	5
Brady	5
Coop	5
Cam	5

9.2 NU Teams Guide to University Owned Vehicles document

1. NU Teams Guide to Vehicle Usage

1.1 Overview

This document is a guideline surrounding the appropriate usage of the University of Newcastle owned vehicles. NU Teams often utilises UON owned vehicles to travel when conducting events off campus, where the process of vehicle [hire](#), and usage can often be overlooked or misunderstood. This guide will allow students to find the specific vehicle they are using, as well as the appropriate procedure surrounding how to use it.

Generally, the most important things to consider when using a UON Vehicle are:

- Full tank of fuel (or close to) on return
- Same level of cleanliness as when picked up
- Logbook completed (app or paper)
- Cost Collector: 1020374

2. University Owned Vehicles

2.1 SELS Vehicles

Logbook: Power App correlating to SELS vehicles, allows usage to be recorded (logbook/kms). QR code found in vehicle or on keys to use.

Fuel cards and E-tag should be equipped in all vehicles.

After Hours Key Drop:



9.3 NU Racing Podium Trophy (3rd Place)



9.4 NU Racing 2025 Budget

Approved Total 2025 Budget	Excel Currently Spent	Asana Currently Spent	Asana Delta	Overall Budget Delta	Predicted Total 2025 Budget						
\$60,000.00	\$57,051.59	60,029.02	\$2,976.43	\$2,216.19	\$6,780.81						
GENERAL											
Department Category	System within category	Price & Shipping	No Float*Factor of imp	Explanation of cost/comments	Link to external website	Purchased	Price of Purchase	Expected Remaining	Purchased Notes	Spending Delta	
ENG2200									Includes requested, approved, purchased, delivered and completed sections as of 23/10/2025 - lakers, shirts (assuming 25 were purchased) \$1191.25	\$	-
Comp Riego	ENG2200	\$ 4,300.00	\$ 2,000.00	\$47 per shirt, \$45 per Teeny, \$30ish for Servo per Team, Approx. \$2446 total		Yes	\$ 2,437.17	\$ 62.83		\$	-
FYP Shirts	Comp Riego	\$ 4,300.00	\$ 4,300.00	Should be \$3800 + GST		Yes	\$ 4,180.00	\$ -		\$	320.00
FYP Shirts	FYP Shirts	\$ 1,000.00	\$ 1,000.00	\$60 pp x 17 members (\$47 pp as of 09/25)		No	\$ 953.00	\$ -	assuming 20 FYP shirts were purchased (\$ paid for extra, 25 2200 shirts, remaining (20) FYP shirts) - (6000 - 25 x \$47)	\$	47.00
Misc. Admin	Calendar	\$ -	\$ -	No cost allocated initially		Yes	\$ 32.75	\$ -		\$	-
Place S&T Repair	END of Month Mug February	\$ -	\$ -	No cost allocated initially		Yes	\$ 6.00	\$ -		\$	6.00
END of Month Mug March	END of Month Mug April	\$ -	\$ -	No cost allocated initially		Yes	\$ 6.00	\$ -		\$	6.00
END of Month Mug May	END of Month Mug June	\$ -	\$ -	No cost allocated initially		Yes	\$ 6.00	\$ -		\$	6.00
END of Month Mug July	END of Month Mug August	\$ -	\$ -	No cost allocated initially		Yes	\$ 6.00	\$ -		\$	6.00
END of Month Mug September	END of Month Mug October	\$ -	\$ -	No cost allocated initially		Yes	\$ 6.00	\$ -		\$	6.00
END of Month Mug November	END of Month Mug December	\$ -	\$ -	No cost allocated initially		Yes	\$ 6.00	\$ -		\$	6.00
Design Booklets	Umbrella	\$ -	\$ -	No cost allocated initially		No	\$ -	\$ 200.00		\$	200.00
General Total	General Total	\$ 7,300.00	\$ 8,000.00				\$ 7,582.92	\$ 288.83		\$	30.25
MECH											
Department Category	System within category	Price & Shipping	No Float*Factor of imp	Explanation of cost/comments	Link to external website	Purchased?	Price of Purchase	Expected Remaining	Purchased Notes	Spending Delta	
M/C4 Power Box	Aluminium brake plate	\$ -	\$ -	Designed and manufactured in house, material free from		Yes	\$ -	\$ -	Calm Aluminium Order: 3546	\$	-
Aluminium motor plate	Chain guard	\$ -	\$ -	Designed and manufactured in house, material free from		Yes	\$ -	\$ -	Calm Aluminium Order: 3546	\$	-
Power Box - Oil	Chain guard	\$ 11,700.00	\$ -	Mild steel chain guard, estimated figure		Yes	\$ -	\$ -	REMAINING	\$	1,234.24
Dreder, FSAE V2 Differential		\$ 5,800.00	\$ 3,700.00	V2 - 10mm including shipping and any special tooling - Float includes 50% for unknown variables	https://autotech.com/30493381-dreder-fsa-e-differential-v2-v3.html	Yes	\$ 4,076.92	\$ -	\$11700 allocated originally, reduced to \$3500, see remaining remaining total adjacent then down, see	\$	5,423.09
Dreder, FSAE Drive Shafts		\$ 1,500.00	\$ 3,000.00	Float includes 50% as no price can be found, will email dreder	https://www.dreder.com/autotech.com/30493381-dreder-fsa-e-differential-v2-v3.html	Yes	\$ 4,130.28	\$ -		\$	1,232.81
Drederhaft Spining Tool		\$ -	\$ -	Includes spline cutting tool for making drive shafts - no initial cost		Yes	\$ 363.08	\$ -		\$	363.73
Powerbox Bearings		\$ -	\$ -	Transmission shaft bearing and dreder bearings: 625-2RS JP, 6210-2RS JP, 6220 DCU - No initial cost		Yes	\$ 79.20	\$ -		\$	850.53
Transmission Shafts		\$ -	\$ -	100mm long @ 100mm OD x 2 (100mm round bar of 4340 300mm long) - No initial cost		Yes	\$ 226.79	\$ -		\$	623.74
Shaft Spinning - Newcastle Ge		\$ -	\$ -	Site visit completed, need to get a quote on spinning and heat treating 2 shafts - No initial cost - \$240 in 2024		Yes	\$ 706.72	\$ -		\$	81.98
Powerbox structure		\$ -	\$ -	Designed and manufactured in house, material free from		Yes	\$ -	\$ -		\$	81.98
Chain Guard		\$ -	\$ -	Designed and manufactured in house, material free from		Yes	\$ -	\$ -		\$	81.98
Dreder Spindles		\$ -	\$ -	No Cost on initial budget		Yes	\$ 291.28	\$ -		\$	473.25
Drederhaft Material		\$ -	\$ -	No Cost on initial budget		Yes	\$ 119.10	\$ -		\$	594.36
Spindle Locking Nuts		\$ -	\$ -	Wheel speed sensor encoder locking nut - No initial cost		Yes	\$ 67.65	\$ -		\$	652.01
Dreder inboard tripod clip bearings & seals		\$ -	\$ -	No initial cost (Thought it would be included in dreder order)		Yes	\$ 38.68	\$ -		\$	751.67
CV Boots for Dreder		\$ -	\$ -	No Cost on initial budget		Yes	\$ 24.00	\$ -		\$	776.67
Dreder spindle end mill (ANFIDP2000TP)		\$ -	\$ -	No initial cost - Supplied from live tools		Yes	\$ 166.65	\$ -		\$	942.22
Differential Oil		\$ -	\$ -	Fluorinated Racing Oil		Yes	\$ 99.00	\$ -		\$	1,041.22
Chain Guide Material		\$ -	\$ -	Black Tape		Yes	\$ 64.86	\$ -		\$	1,106.25
Dreder Sprocket Adapter Circle		\$ -	\$ -	No initial cost (Thought it would be included in dreder order)	https://www.bonhant.com.au/50mm-stainless-4140-external-circle-dn471	Yes	\$ 19.00	\$ -		\$	1,125.25
Spine Dreder Housing Bearings		\$ -	\$ -	No Cost on initial budget		Yes	\$ 108.26	\$ -		\$	1,234.24
TBD		\$ -	\$ -			No	\$ -	\$ -		\$	1,234.24
TBD		\$ -	\$ -			No	\$ -	\$ -		\$	1,234.24
Solar Panel Hub		\$ 186.40	\$ -			No	\$ -	\$ -		\$	1,234.24
Driver Sprocket		\$ 21.00	\$ 25.20	Float includes 20%	https://dpelartsuperstore.com/product/8-as-tech-400c	No	\$ -	\$ -		\$	25.20
Driven sprocket		\$ 21.00	\$ 25.20	Float includes 20%	https://dpelartsuperstore.com/product/8-as-tech-engine-sprocket-13x2-400c-100-bd-20-250-400c	No	\$ -	\$ -		\$	25.20
Tune up Engine work		\$ 40.00	\$ 48.00	Spark plug, carb clean, etc	https://dpelartsuperstore.com/product/8-as-tech-engine-sprocket-13x2-400c-100-bd-20-250-400c	No	\$ -	\$ -		\$	48.00
Brake attention		\$ 80.00	\$ 89.00	Float includes 10%	https://dpelartsuperstore.com/product/8-as-tech-engine-sprocket-13x2-400c-100-bd-20-250-400c	No	\$ -	\$ -		\$	89.00
N/16 Flatbush		\$ 936.00	\$ -	Original Allocation - Bearings, Service & Brakes		No	\$ -	\$ -		\$	221.52
Drive shafts		\$ -	\$ -	No Cost on initial budget		Yes	\$ 119.10	\$ -		\$	119.10
Bearings		\$ -	\$ -	No Cost on initial budget		Yes	\$ 186.95	\$ -		\$	186.95
Service Parts		\$ 700.00	\$ 840.00	Fluents, bearings, consumables, etc - Float includes 20%		Yes	\$ 84.00	\$ -		\$	675.40
Brake attention		\$ 80.00	\$ 96.00	Brake rebuild flush/ refurb - Float includes 20%		Yes	\$ 186.95	\$ -		\$	30.95
Fuel Pump		\$ -	\$ -	Replacement eBay fuel pump		Yes	\$ 37.69	\$ -		\$	37.69
Barbs for N/16 fuel system		\$ -	\$ -	No Cost on initial budget		Yes	\$ 31.14	\$ -	\$336.00 Allocated originally, \$632.49 spent during semester 1. See remaining total adjacent	\$	287.37
More barbs for N/16 fuel tank		\$ -	\$ -	No Cost on initial budget		Yes	\$ 29.21	\$ -		\$	267.56
Facilities		\$ -	\$ -	No Cost on initial budget		No	\$ -	\$ -		\$	237.35
Yield on fuel bar		\$ -	\$ -	No Cost on initial budget		Yes	\$ 16.04	\$ -		\$	251.02
Thread Sealant		\$ -	\$ -	No Cost on initial budget		No	\$ -	\$ 30.00		\$	221.02